Ankit Kr Sinha

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.08
K.M Academy	Class XII (CBSE)	2019	80.40%
DAV Public School	Class X (CBSE)	2017	9.80%

WORK EXPERIENCE

A.P. Moller - Maersk

Software Engineer Intern

- Implemented configurations and code migrations utilizing .NET 6 for the ADCv2 release, reducing VM build time by ~30%
- Contributed to the adoption of Kubernetes-based deployment orchestration, optimizing application deployment workflows.
- Implemented runtime-generated DLL protection during the build process.
- Managed live ad-hoc configuration changes across 130+ vessel connectors, maintaining configuration consistency with requirements from ship engineers.
- Technologies used: C#, .NET 6, Kubernetes, Docker, MySQL, Git, Python, Grafana.

Omdena

Jr. ML Engineer

- Developed multiple NLP models to identify and categorize war-crime-related posts and deployed an MVP web application for visualizing reports with automatic updates from incoming data.
- Collected social-media data from Reddit and Twitter, defining 11 war-crime categories from academic criteria and manually labeling posts to create two datasets for model development.

SKILLS

- **Programming Languages:** C++, Python, Java, C#
- **Backend:** .NET 6, Express.js, Node.js, Flask | **Database:** MySQL, PostgreSQL, MongoDB, Redis
- **Systems & Tools:** Docker, Kubernetes, Git, Jenkins, Linux, LLVM, CUDA, Grafana
- **ML & AI:** RAG, Scikit-Learn, Tensorflow, Agentic AI, NLP, LangChain, LangGraph

PROJECTS

Domain-Specific Text-to-SQL System for GST Analytics

- Built a Text-to-SQL pipeline with XiYanSQL-QwenCoder-7B, vLLM, PostgreSQL & SQLAlchemy, using dynamic M-Schema grounding, SQL validation & controlled execution with CTE-aware parsing, SELECT-only enforcement, table allowlisting, read-only DB, timeouts and result limits
- Implemented dual-channel RAG (BGE-large, FAISS and BM25/RRF) with hybrid few-shot retrieval and value-grounded schema retrieval; FK-ancestor closure and core-table injection improved required-table coverage from 71.5% to 98.3%.

LLVM-Guided CUDA Kernel Fission & Fusion

- Developed a C++ LLVM compiler analysis pass to analyze CUDA LLVM IR using instruction/phase classification, def-use dependency analysis, memory-access patterns, register-pressure & occupancy modeling, and a cost model to identify profitable kernel fusion/fission opportunities.
- Implemented IR-level kernel transformations with legality/dependency checks and PTX generation via LLVM NVPTX, validating correctness and predicted-vs-measured performance using ptxas, NVIDIA Nsight Compute/Systems, and Rodinia/benchmark kernels.

Automated Cheque Verification & Clearing System

- Developed an automated cheque verification pipeline using image preprocessing, template-based segmentation, Tesseract OCR and CNN-based transfer learning for extracting printed fields and recognizing handwritten digits, achieving 99.1% test accuracy.
- Implemented SIFT + SVM-based signature verification using geometric and invariant features, achieving 98.1% accuracy on the IDRBT cheque dataset.

TransactiWar | Secure Banking Web Application

- Developed a PHP-MySQL banking platform with session-based authentication, CSRF token validation, password hashing, IP-based login throttling, prepared SQL statements, and secure session/cookie configuration.
- Implemented ACID-compliant fund transfers with transactional rollback, user/profile management, transaction history, activity auditing, secure file handling, HTTP security headers, and Dockerized/Apache deployment.

SCHOLASTIC ACHIEVEMENTS

Competitive Programming: Coding Ninjas **2235** | CodeChef **1604 (3 star)** max rating | Codeforces **1315** max rating

Selected for Amazon ML Summer School 2022, a competitive program covering machine learning concepts and applications.

IEEE Best Project Award, Open House 2023, for research project *"Automated Cheque Verification and Clearing System"*

AIR 681 in AccioWars Coding Contest by Accio Jobs with 12k+ active participants.

POSITIONS OF RESPONSIBILITY

Placement Coordinator -- Office of Career Services (OCS), IIT Hyderabad

Coordinated between recruiters, students, and OCS to facilitate smooth internship and placement processes.